

Convergence of the Discrete Consensus-Based Optimization Algorithm with Random Batch Interactions

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Overview

- 1 Introduction; Consensus-Based Optimization
- 2 Random Batch Method (RBM) and its use on CBO
- 3 Previous works; kinetic equation for CBO
- 4 Main analysis; Contraction of diameter for random network
- 5 Summary

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consensus-based optimization (CBO)

CBO: an evolutionary type of **gradient-free optimization algorithms** to find the minimum of a given cost function.

Similar methods; **Ant Colony Optimization, Particle Swarm Optimization, Wolf-pack optimization, Genetic algorithm, etc.:**

- 1 First, **spread the particles** into the domain.
- 2 Second, **evaluate current values** from particles' positions.
- 3 Third, process time-**evolution toward the possible minimum** positions.

For a given $L(x)$, we want $x_i(t)$ to approach $x_* := \operatorname{argmin}_{x \in \mathbb{R}^d} L(x)$.

consensus-based optimization (CBO)

SIMULATIONS:

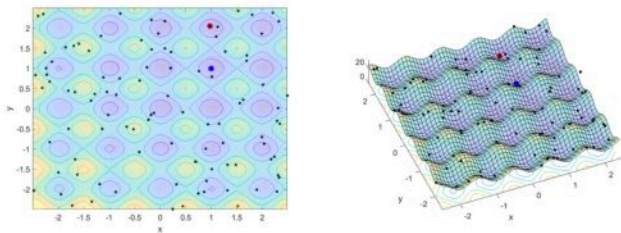


Figure: Initial particle distribution and the Rastrigin cost function

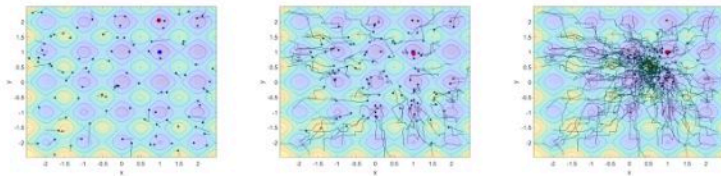


Figure: Particle distribution at (left) $t = 2$, (middle) $t = 10$, (right) $t = 50$

consensus-based optimization (CBO)

Therefore, each particle **explores the domain**:

- $x_i = i$ -th agent's guess for $\operatorname{argmin}_{x \in \mathbb{R}^d} L(x)$
- Iterate on $t \in \mathbb{N}$,

$$x_i(t+1) = x_i(t) + (\text{interaction with other } x_j(t)\text{'s}), \quad i = 1, \dots, N$$

Three questions:

- 1 **[Consensus]** $x_i(t) - x_j(t)$ decays to zero.
- 2 **[Convergence]** all $x_i(t)$ converge to its limit $x_i(\infty)$.
- 3 **[Optimality]** $x_i(\infty) \approx \operatorname{argmin}_{x \in \mathbb{R}^d} L(x)$ for some i .

consensus-based optimization (CBO)

CBO is a kind of simplified version of multi-point optimization methods.

Algorithm [K.–Ha–Jin–Kim 2022] based on [Carrillo–Jin–Li–Zhu 2021]

$$X_{(t+1)}^i = X_t^i + \gamma(\bar{X}_t^{i,*} - X_t^i) + \text{diag}(\eta_t^{i,1}, \dots, \eta_t^{i,d})(\bar{X}_t^{i,*} - X_t^i),$$

$$\gamma > 0, \quad \eta_t^{i,\ell} \sim \mathcal{N}(0, \sqrt{\zeta}) \quad \text{for each } i, \ell, t \quad \text{and}$$

$$\bar{X}_t^{i,*} := \operatorname{argmin}_{x \in \{X_t^j | j \in N_i(t)\}} L(x), \quad N_i(t) \subset \{1, 2, \dots, N\}.$$

This time-evolution corresponds to a continuous stochastic dynamics:

$$dX_t^i = \lambda(\bar{X}_t^{i,*} - X_t^i)dt + \sigma \text{diag}(\bar{X}_t^{i,*} - X_t^i)dW_t^i,$$

which is surely **gradient-free**.

Numerical simulations

- A common way to set the target point $\bar{X}_t^{i,*}$ of the i -th particle is to set the best point among the whole particles:

$$\bar{X}_t^{i,*} := \operatorname{argmin}_{x \in \{X_t^j\}} L(x), \quad \text{i.e.,} \quad L(\bar{X}_t^{i,*}) = \min_{j=1,\dots,N} L(X_t^j).$$

- The simulation ($N = 100, \gamma = 0.01, \zeta = 0.5$) on **Rastrigin functions** suggests the following probabilities to find the minimum:
In 2D and 3D, nearly 100%, however, more than 7D, less than 50%.

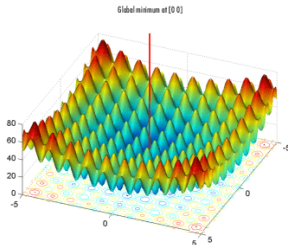


Figure: The landscape of Rastrigin function in $d=2$ (2D).

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Random Batch Method (RBM)

Suggestion on raising success rate: consider complex interaction structure, such as RBM (in [Carrillo–Jin–Li–Zhu 2021]).

Random Batch Method [Jin–Li–Liu, 2020] suggests a random sampling on the interactions at each time-step.

At each time n , we choose (randomly) a partition of $\{1, \dots, N\}$ with size P , ($2 \leq P \leq N$): for $m = \lceil N/P \rceil$,

$$\{1, \dots, N\} = \mathcal{B}_1^n \cup \dots \cup \mathcal{B}_m^n, \quad |\mathcal{B}_\ell^n| = P \quad \text{for } \ell < m \quad \text{and} \quad |\mathcal{B}_m^n| \leq P.$$

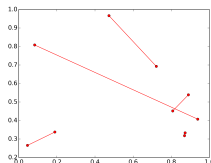


Figure: Pairing from 10 particles ($P = 2$)

Network structure of RBM

From RBM, we the interaction network becomes **sparse**:

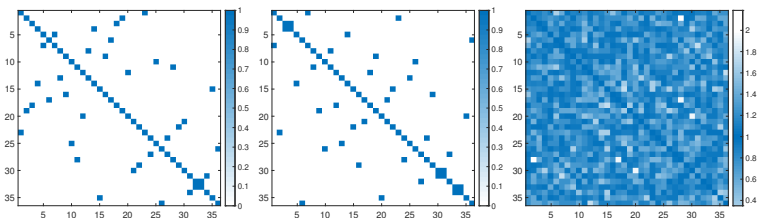


Figure: Color representation of the adjacency matrix.

(Left two) Examples of random networks from RBM ($P = 2$, $N = 36$).

(Right) An average of independent 400 matrices.

Convergence theorem [Jin–Li–Liu, 2020]

Following the law of large numbers, the solution of **RBM converges to the original (all-to-all) dynamics** as the time-discretization goes zero.

Numerical simulations

The simulation result shows that if P gets smaller,
then the success rate grows but the cost of computation also grows.

Success rate	Full batch ($P = 100$)	$P = 50$	$P = 10$
$d = 2$	1.000	1.000	1.000
$d = 3$	0.988	0.983	0.998
$d = 4$	0.798	0.920	0.988
$d = 5$	0.712	0.658	0.931
$d = 6$	0.513	0.655	0.880
$d = 7$	0.388	0.464	0.854
$d = 8$	0.264	0.389	0.832
$d = 9$	0.170	0.323	0.868
$d = 10$	0.117	0.274	0.886

Figure: Success rates from 1000 simulations.

Numerical simulations

The accuracy of algorithm directly depends on the computation time.

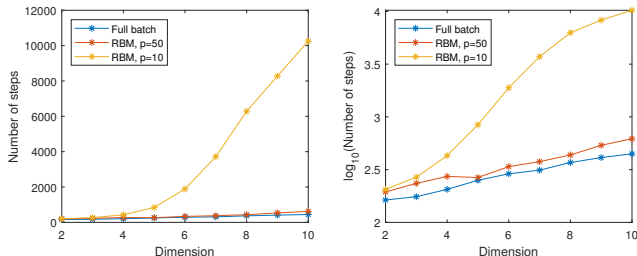


Figure: Average number of steps until stopping criterion holds.

The stopping criterion is made with the change of positions,

$$\sum_{i=1}^N |x_{n+1}^i - x_n^i|^2 < 10^{-3}.$$

consensus-based optimization (CBO)

Algorithm [K.–Ha–Jin–Kim 2022] based on [Carrillo–Jin–Li–Zhu 2021]

$$X_{(t+1)}^i = X_t^i + \gamma(\bar{X}_t^{i,*} - X_t^i) + \text{diag}(\eta_t^{i,1}, \dots, \eta_t^{i,d})(\bar{X}_t^{i,*} - X_t^i),$$

$$\gamma > 0, \quad \eta_t^{i,\ell} \sim \mathcal{N}(0, \sqrt{\zeta}) \quad \text{for each } i, \ell, t \quad \text{and}$$

$$\bar{X}_t^{i,*} := \operatorname{argmin}_{x \in \{X_t^j | j \in N_i(t)\}} L(x), \quad N_i(t) \text{ determined by RBM.}$$

It corresponds to a continuous SDE:

$$dX_t^i = \lambda(\bar{X}_t^{i,*} - X_t^i)dt + \sigma \text{diag}(\bar{X}_t^{i,*} - X_t^i)dW_t^i.$$

Here $\bar{X}_t^{i,*}$ is the minimum guess from the batch of the i -th particle.

Objectives: here we want to guarantee the termination of the algorithm by showing convergence of the particles: $X_t^i \rightarrow X_\infty$ as $t \rightarrow \infty$.

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Previous works

Proposal of algorithms & Analysis of the convergence

- [Askari-Sichani–Jalili 2013] proposed and analyzed CBO without noise
- [Pinnau–Totzeck–Tse–Martin 2017] proposed CBO with noise
- [Carrillo–Choi–Totzeck–Tse 2018] analyzed the convergence of the kinetic CBO dynamics
- [Ha–Jin–Kim 2020,2021] analyzed the convergence of a simplified CBO algorithm.
- [Fornasier–Huang–Pareschi–Sünnen 2020] proposed CBO on hypersurfaces
- [Kim–Kang–Kim–Ha–Yang 2020] proposed CBO on the Stiefel manifold
- [Carrillo–Jin–Li–Zhu 2021] proposed CBO for high dimensional problems

Two examples in Literature

The exploration of CBO algorithm has two parts of randomness.

Algorithm with interaction network [Askari-Sichani–Jalili 2013]

$$x_i(t+1) = x_i(t) + \gamma(\bar{x}_i^*(t) - x_i(t)), \quad \bar{x}_i^*(t) = \operatorname{argmin}_{x_k(t): k \in N_i(t)} L(\cdot)$$

Algorithm for noisy trajectory [Pinnau–Totzeck–Tse–Martin 2017]

$$dX_t^i = \lambda(\bar{X}_t^* - X_t^i)dt + \sigma|\bar{X}_t^* - X_t^i|dW_t^i,$$

with

$$\bar{X}_t^* := \frac{1}{\sum_{j=1}^N e^{-\beta L(X_t^j)}} \sum_{j=1}^N e^{-\beta L(X_t^j)} X_t^j.$$

The second $\bar{X}_t^*$ is from the **Laplace principle**, which converges to the argument minimum as $\beta \rightarrow \infty$. This is an **observable** in F-P dynamics.

Algorithm in [Pinnau–Totzeck–Tse–Martin 2017]

Algorithm [Pinnau–Totzeck–Tse–Martin 2017]

$$dX_t^i = \lambda(\bar{X}_t^* - X_t^i)dt + \sigma|\bar{X}_t^* - X_t^i|dW_t^i,$$

with

$$\bar{X}_t^* := \frac{1}{\sum_{j=1}^N e^{-\beta L(X_t^j)}} \sum_{j=1}^N e^{-\beta L(X_t^j)} X_t^j.$$

We can formally send $N \rightarrow \infty$ to get

$$\bar{X}_t^* \rightarrow \frac{1}{\int_{\mathbb{R}^d} e^{-\beta L(x)} d\rho_t} \int_{\mathbb{R}^d} e^{-\beta L(x)} x d\rho_t, \quad \rho_t : \text{prob. measure of } X_t^i.$$

If L has a **unique minimizer x_* in the support** of ρ_t , then

$$m[\rho_t] := \frac{1}{\int_{\mathbb{R}^d} e^{-\beta L(x)} d\rho_t} \int_{\mathbb{R}^d} e^{-\beta L(x)} x d\rho_t \rightarrow x_* \quad \text{as } \beta \rightarrow \infty.$$

Mean-field limit to the kinetic dynamics

From the mean-field limit process, the dynamics of

$$dX_t^i = \lambda(\bar{X}_t^* - X_t^i)dt + \sigma|\bar{X}_t^* - X_t^i|dW_t^i$$

becomes dynamics of the density $\rho_t \in \mathcal{P}(\mathbb{R}^d)$ as a Fokker-Planck equation:

$$\partial_t \rho_t = \lambda \nabla \cdot ((x - m[\rho_t])\rho_t) + \frac{\sigma^2}{2} \Delta(|x - m[\rho_t]|^2 \rho_t).$$

Theorem (Convergence) [Pinnau–Totzeck–Tse–Martin 2017]

If λ is large enough (compared to d , σ^2 , and $e^{-\beta}$), then $\mathbb{E}(\rho_t)$ converges and

$$\text{Var}(\rho_t) = O(e^{-ct}), \quad t \rightarrow \infty.$$

Idea: $\frac{d}{dt} \text{Var}(\rho_t) = -2\lambda \text{Var}(\rho_t) + (d\sigma^2/2) \int (x - m[\rho_t])^2 d\rho_t.$

Limitations of kinetic approach

CBO with RBM converges to the kinetic equation as $\tau \rightarrow 0$ and $N \rightarrow \infty$. However, kinetic equation limit has **two significant limitation**.

1 Dynamics ignores network structure:

The particles are not-distinguishable; all-to-all interactions happen.

2 Sample space covers the whole space:

The whole values we need; **we already know the global minimum**.

[Fornasier–Klock–Riedl, 2021]

We should get back to the **discrete-time dynamics** in order to analyze the performance of algorithms.

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Main result: Consensus of a general CBO algorithm

For the CBO algorithm

$$X_{(t+1)}^i = X_t^i + \gamma(\bar{X}_t^{i,*} - X_t^i) + \text{diag}(\eta_t^{i,1}, \dots, \eta_t^{i,d})(\bar{X}_t^{i,*} - X_t^i),$$

we assume $\bar{X}_t^{i,*}$ is in the convex hull of information at i :

$$\bar{X}_t^{i,*} := \sum_{j \in N_i(t)} f_{ij}(t) X_t^j : \text{convex combination}, \quad \sum_{j \in N_i(t)} f_{ij}(t) = 1, \quad \forall i.$$

Theorem (K.-Ha-Jin-Kim 2022)

For sufficiently small (not depending on d) $\zeta := \text{Var}(\eta_t^{i,l})$,
(1) for some positive random variable ε ,

$$\mathbb{E} \max_{i,j} \|X_t^i - X_t^j\| = \mathcal{O}(e^{-\varepsilon t}), \quad t \rightarrow \infty.$$

(2) the following holds almost surely: for some positive random ε ,

$$\max_{i,j} \|X_t^i - X_t^j\| = \mathcal{O}(e^{-\varepsilon t}), \quad t \rightarrow \infty.$$

Algorithm in [Askari-Sichani–Jalili 2013]

We may **rewrite the dynamics** without noise in [Askari-Sichani–Jalili 2013],

$$x_i(t+1) = x_i(t) + \gamma(\bar{x}_i^*(t) - x_i(t)), \quad \bar{x}_i^*(t) = \operatorname{argmin}_{x_k(t): k \in N_i(t)} L(\cdot),$$

as in the **matrix form**:

$$X(t+1) = A(t)X(t).$$

For example, if there are 4 particles and the third is the $\bar{x}_i^*(t)$ for all i :

$$A(t) = \begin{bmatrix} 1-\gamma & 0 & \gamma & 0 \\ 0 & 1-\gamma & \gamma & 0 \\ 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & \gamma & 1-\gamma \end{bmatrix}.$$

In this case, we have

$$|x_i(t+1) - x_j(t+1)| \leq (1-\gamma)|x_i(t) - x_j(t)|.$$

Ergodicity coefficient

Question: How can we prove the consensus?

Answer: Analysis on the **diameter** as the variance in the kinetic equation.

$$\mathcal{D}(x) := \max_{i,j} |x_i - x_j|.$$

For a stochastic matrix, define **ergodicity coefficient** as

$$\alpha(A) := \min_{i,j} \sum_k \min\{a_{ik}, a_{jk}\} \in [0, 1].$$

Note: $\alpha(A) = 1 \Leftrightarrow$ all rows of A are identical.

Proposition [Askari-Sichani–Jalili 2013] from [Markov 1906]

$$\mathcal{D}(Ax) \leq (1 - \alpha(A))\mathcal{D}(x).$$

Problem 1; the case of ergodicity zero

For example, the stochastic matrix may look like

$$A = \begin{bmatrix} 1-\gamma & \gamma & 0 & 0 \\ 0 & 1-\gamma & \gamma & 0 \\ 0 & 0 & 1-\gamma & \gamma \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{such as} \quad A = \begin{bmatrix} 0.9 & 0.1 & 0 & 0 \\ 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.9 & 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Since this is a block matrix, the ergodicity constant should be zero:

$$\alpha(A) := \min_{i,j} \sum_k \min\{a_{ik}, a_{jk}\} = 0. \quad (i=1, j=3)$$

However, the network is connected; if we consider A^2 and A^3 ,

$$A^2 = \begin{bmatrix} 0.81 & 0.18 & 0.01 & 0 \\ 0 & 0.81 & 0.18 & 0.01 \\ 0 & 0 & 0.81 & 0.19 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad A^3 = \begin{bmatrix} 0.729 & 0.243 & 0.027 & 0.001 \\ 0 & 0.729 & 0.243 & 0.028 \\ 0 & 0 & 0.729 & 0.271 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Therefore, $\alpha(A^3) \neq 0$ since $i=1$ and $j=4$ has length 3 connectivity.

Analytical result in [Askari-Sichani–Jalili 2013]

Therefore, if the network is **connect** so that

$$\lim_{n \rightarrow \infty} \alpha(A^n) = 1,$$

then $\mathcal{D}(A^n x) \leq (1 - \alpha(A^n))\mathcal{D}(x) \rightarrow 0$.

It is convenient to check the convergence to 1 as follows. Define

$$A((t, s]) := A(t-1)A(t-2) \dots A(s) \quad (t > s).$$

Proposition [Askari-Sichani–Jalili 2013]

Assume that there exists $0 = t_0 < t_1 < t_2 < \dots$ satisfying

$$\sum_{i=1}^{\infty} \alpha(A((t_i, t_{i-1}])) = \infty.$$

Then $\lim_{n \rightarrow \infty} \alpha(A^n) = 1$ and $\mathcal{D}(X(t)) \rightarrow 0$ as $t \rightarrow \infty$.

Problem 2; connectivity of RBM

The problem occurs when $\bar{x}_i^* \neq \bar{x}_j^*$.

For example, the stochastic matrix may look like

$$A(t) = \begin{bmatrix} 1-\gamma & \gamma & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1-\gamma & \gamma \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Since this is a block matrix, the ergodicity constant should be zero:

$$\alpha(A) := \min_{i,j} \sum_k \min\{a_{ik}, a_{jk}\} = 0. \quad (i=1, j=3)$$

Therefore, the random network needs connectivity.

Lemma: positive ergodicity of a network

Assume that for any i and j , suppose that **the information of i at time t is observed by j at time $t+m$** . Then, $\alpha(A((s+m, s])) \geq \gamma(1-\gamma)^m$.

Consensus with RBM

Now we may prove that the diameter decays almost surely.

First, from **ergodicity** argument, we have

$$\mathcal{D}(X(t_{km})) \leq \exp\left(-\gamma(1-\gamma)^m \sum_{s=1}^k \mathcal{G}_{s,m}\right) \mathcal{D}(X(0)),$$

where $\mathcal{G}_{s,m}$ is 1 if all particles are connected by length m , and 0 otherwise. (This is a **random variable** with positive probability to be 1)

Note that

$$\lim_{k \rightarrow \infty} \frac{1}{k} \sum_{s=1}^k \mathcal{G}_{s,m} = \mathbb{E}[\mathcal{G}_{s,m}] = p_m > 0.$$

Therefore, we conclude the decay of the diameter

$$\begin{aligned} \mathcal{D}(X(t_{km})) &\leq \exp(-\Lambda(m, k)k) \mathcal{D}(X(0)), \\ \lim_{k \rightarrow \infty} \Lambda(m, k) &= \gamma(1-\gamma)^m p_m. \end{aligned}$$

In other words, **the algorithm satisfy consensus.**

Ergodicity with noise?

Now we consider noise;

$$X_{(t+1)}^i = X_t^i + \gamma(\bar{X}_t^{i,*} - X_t^i) + \text{diag}(\eta_t^{i,1}, \dots, \eta_t^{i,d})(\bar{X}_t^{i,*} - X_t^i),$$

Then, the network structure

$$A(t) = \begin{bmatrix} 1 - \gamma & \gamma & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 - \gamma & \gamma \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

becomes

$$A_\varepsilon(t) = \begin{bmatrix} 1 - \gamma + \varepsilon & \gamma - \varepsilon & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 - \gamma + \varepsilon & \gamma - \varepsilon \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Then, A is **stochastic** but can have **negative ergodicity**.

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Summary and remaining questions

Remarks

- 1** The convergence of CBO holds with random batch method.
- 2** This is the first result on the convergence of CBO with individual random motion on particles.
- 3** No performance result yet on the global minimizer.

Thank you very much